

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. FEATURE SUMMARY:

The ComicCraft platform is equipped with an array of interconnected functional features designed to completely abstract the complexities of Generative AI prompting and document formatting from the end-user.

3. COMPREHENSIVE FEATURE BREAKDOWN

- **Feature 1: Dynamic User Input Configuration**
 - *Description:* An intuitive HTML frontend interface that captures specific narrative variables (Idea, Main Character) and aesthetic parameters (Art Style, Genre) without requiring complex prompt engineering from the user.
- **Feature 2: Sequential Narrative Structuring**
 - *Description:* Utilizes the high-speed Google Gemini 1.5 Flash model to dissect the user's prompt into a strict, logical JSON array, ensuring the story flows coherently across exactly 5 distinct scenes.
- **Feature 3: Contextual Dialogue Generation**
 - *Description:* Employs Google Gemini 1.5 Pro to expand upon the structural JSON, generating engaging conversational dialogues and narration tailored to the specific genre selected by the user.
- **Feature 4: Autonomous Image Synthesis**
 - *Description:* Integrates directly with the Hugging Face Stable Diffusion API to autonomously generate, retrieve, and locally save high-resolution comic illustrations based on the AI-written scene descriptors.
- **Feature 5: Autonomous Error Recovery System**
 - *Description:* Incorporates a highly resilient backend fail-safe that detects API timeouts or network blocks. It instantly deploys a

localized Python PIL script to generate graphic placeholders, guaranteeing the application never experiences a fatal crash.

- **Feature 6: Interactive Digital Preview**

- *Description:* Employs the Jinja2 templating engine to dynamically parse the backend dictionary data, rendering an interactive, easy-to-read comic strip preview directly within the web browser.

- **Feature 7: Programmatic Document Compiler**

- *Description:* Integrates the Python FPDF module to algorithmically calculate X and Y coordinates, stitching the generated .jpg assets and text strings into a perfectly aligned, offline-ready PDF document for immediate download.